



9/1/2026

N + - Network Fundamentals & Building Networks.

DNS Server Configuration.

TASK 1.

- ❖ Install and set up a DNS server on your local machine or a VM using BIND (Linux) or Simple DNS Plus (Windows), then create a new DNS zone for the domain musicmania.in.

➤ DNS Server and Zone Setup:

Steps:

- Install and configure BIND DNS Server on a Linux machine or VM.
- Create a new DNS zone named musicmania.in.
- Add the required zone configuration and save the settings.
- Restart the DNS service to apply the changes.

Result:

The DNS server was configured successfully, and a new DNS zone for musicmania.in was created.

TASK 2.

- ❖ Add an A record in your DNS server configuration to map **www.musicmania.in** to the IP address **192.168.1.50**, then use the **nslookup** or **dig** command to verify the mapping works.

➤ Add and Verify A Record:

A Record

www.musicmania.in → 192.168.1.50

Verification Command

nslookup www.musicmania.in

Result:

The *nslookup* command returned **192.168.1.50**, confirming that the A record was configured correctly.

TASK 3.

❖ **Simulate a typo by adding a CNAME record for `play.musicmania.in` that points to `www.musicmania.in`, then test if accessing `play.musicmania.in` resolves to the correct IP address.**

➤ **CNAME Record Configuration:**

CNAME Record

play.musicmania.in → *www.musicmania.in*

Verification

nslookup play.musicmania.in

Result:

The lookup resolved `play.musicmania.in` to **192.168.1.50**, confirming that the CNAME record works correctly.

TASK 4.

- ❖ **Simulate Your DNS server is not resolving the domain musicmania.in as expected. Document the step-by-step troubleshooting process you would follow, including checking configuration files, restarting the DNS service, and testing with nslookup or dig. Include at least three different troubleshooting commands or checks in your documentation.**

➤ DNS Troubleshooting:

Troubleshooting Steps:

Check the DNS configuration files to find any incorrect zone or record settings.

Check the DNS service status using:

```
systemctl status bind9
```

Restart the DNS service to apply the correct configuration:

```
sudo systemctl restart bind9
```

Test the domain resolution using:

```
nslookup musicmania.in
```

Verify the DNS response using:

```
dig musicmania.in
```

Result:

These checks help identify configuration errors, restart the DNS service, and confirm whether **musicmania.in** is resolving correctly.